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COSC320-001

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Lab 11 ReadMe

Summary:

I started off with the previous code created in Lab 10, then I reviewed the notes for DFS, and topological sort to create the other functions for this lab. This lab I had to use more of the standard library functions such as map, stack, queue, and list.

Time Complexity:

- Depth First Search is: $O(|V| + |E|)$.
- Topological Sort is: $O(|V| + |E|)$ too.
- Strongly Connected Component is: $\Theta(|V| + |E|)$ too.

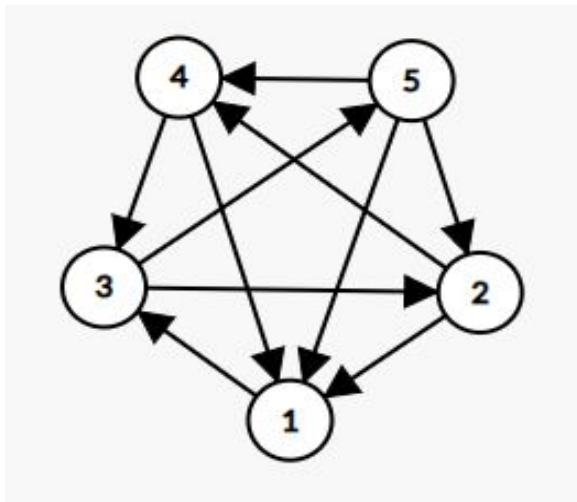
Code Improvements:

Fixing the SCGC (Strongly connected graph component) function would be good. (At least I believe it is broken).

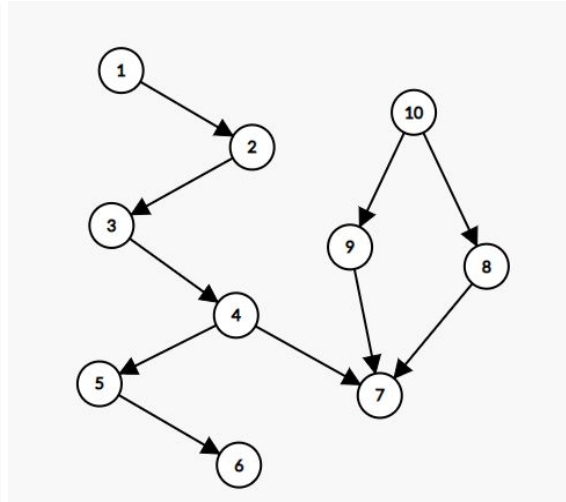
It could also be templated, however; ain't nobody got time for that!

Test graphs used:

Test 1



Test 2



-Created using: https://csacademy.com/app/graph_editor/